

Concept Review

Infrared Distance

Why Do We Need Infrared Distance Sensors?

When designing an electromechanical system, we consider electrical requirements, mechanical complexity and environmental impact on the sensor's measurement integrity. Infrared distance sensors emit infrared pulses and measure the reflected infrared energy. Mechanically, they depend on a lens to focus light for both the emitter and detector. Infrared distance sensors can be found in robotics, aviation, smartphones, etc.

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Using the same concept as Time-of-Flight sensors, infrared (IR) distance sensors use emitters and receivers to convert infrared radiation into an analog distance measurement. Figure 1. shows examples of different models/types of IR distance sensors.



Sharp GP2Y0A21YK0F



Infrared Pair Tube

Figure 1: Different configurations for infrared distance sensors.

Figure 2 shows an example of a laser-based IR distance sensor which estimates the straight-line distance to a subject of interest.

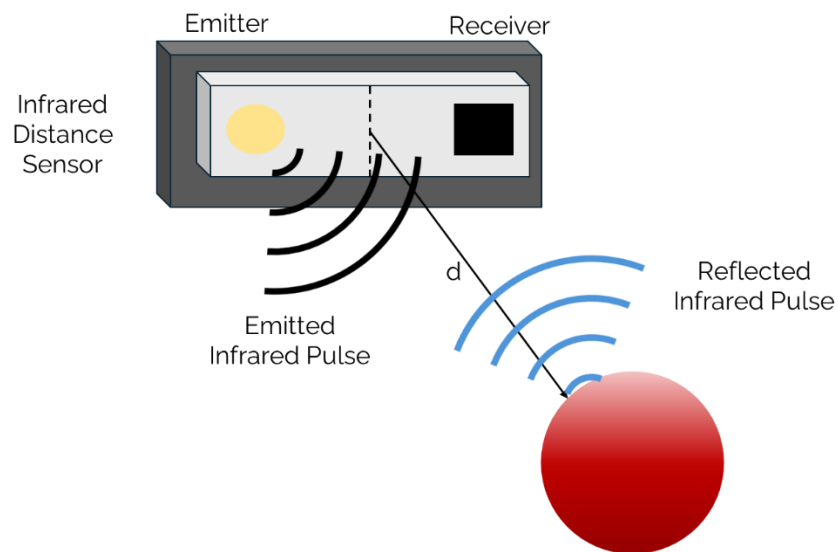


Figure 2: Distance estimate from an Infrared Distance sensor

Sound focused distance sensors, such as ultrasonic sensors, use elements such as acoustic cones to focus sound. IR distance sensors operate similarly but instead use lenses to focus the reflected infrared wave.

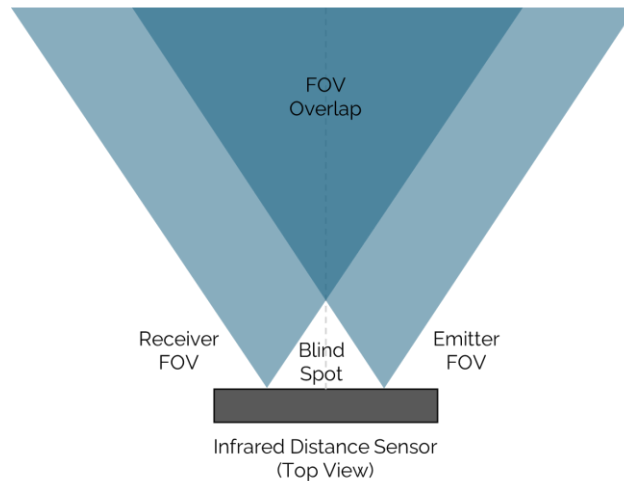


Figure 2: Lens field of view overlap.

With the use of lenses to focus pulsed infrared waves, distance measurements have a working range. As an example, for the Sharp GP2Y0A21YK0F the valid distance measurement range is from 10cm up to 80 cm. Another important characteristic of infrared distance sensors is the effect of infrared energy dissipation as a function of distance. As an example, Figure 3. shows the relationship between distance to a reflective object vs Output Voltage for the Sharp GP2Y0A21YK0F.

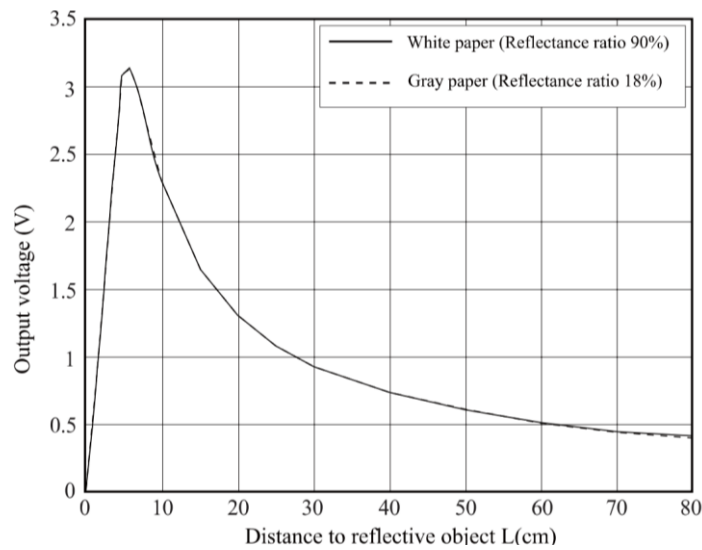


Figure 3. Distance to reflective object vs Output voltage

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